

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch: Mechanical Engineering Semester:VI Subject Code & Name: BTMOE505B Renewable Energy Sources Max Marks: 60 Date:14/02/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain renewable energy sources.	CO1	6
B)	Explain about fossil fuel & nuclear power.	CO1	6
C)	Explain Nuclear Fission and Fusion.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	Explain concept of solar energy with their applications.	CO2	6
B)	Explain with neat sketch construction & working of sun shine recorder.	CO2	6
C)	Explain solar Geometry With neat sketch.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Explain difference between Non-Concentrating and Concentrating type Solar Collectors.	CO2	6
B)	Explain with neat sketch construction of flat plate collectors & list out its limitation.	CO2	6
C)	Explain with neat sketch Heliostat concentrators & list out its advantages.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	Explain with neat sketch natural circulation water heating system	CO3	6
B)	Explain construction working of photo voltaic cell.	CO3	6
C)	Explain Solar drying system with advantages & disadvantages.	CO3	6
Q. 5	Solve Any Two of the following.		12

A)	What is wind energy? Give classification of types of wind mills.	CO4	6
B)	What is biomass energy? Explain with neat sketch floating drum type biogas plant.	CO4	6
C)	Write a short note on Geothermal energy & OTEC system	CO4	6
*** End ***			

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